

NTC_CC105 - Complexity Analysis

Smart Campus Student Resource Management System | BSIT 2.4 | Instructor: Darwin Vargas

Big-O Analysis

Operation	Time Complexity	Explanation
Direct array access	$O(1)$	Accessing a known array index takes constant time.
Student array traversal	$O(n)$	Each occupied student record is visited once.
Linear student search	Best $O(1)$; Worst $O(n)$	The target may be first, or the search may examine every stored record.
Student insertion	$O(n)$	Duplicate-ID validation uses a linear scan before insertion.
Student update	$O(n)$	The program first finds the Student ID using linear search.
Student deletion	$O(n)$	The record is found and following elements may need to shift left.
4 x 4 / n x n matrix traversal	$O(n^2)$	Nested loops visit each matrix cell.
Row/column utilization totals	$O(n^2)$	All matrix cells are processed while accumulating totals.

Required core complexities: direct array access $O(1)$, traversal $O(n)$, linear search $O(n)$, and n x n matrix traversal $O(n^2)$.